Case Study: U.S. Shoe Manufacturing (McKinsey Style)

Client

A leading **U.S. shoe manufacturer** currently produces its entire product line domestically. Rising labor costs and competitive pressures have pushed the company to consider **offshoring some or all production**.

Objective

Determine whether the client should offshore manufacturing, where to offshore, and what % of shoes should be produced domestically vs. offshore.

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1. Framework: Factors to Consider

- **Demand Factors**: volatility, growth, diversity, competition, service level
- **Technology Factors**: access to skilled labor, infrastructure, innovation
- **Supply Factors**: lead time, supplier availability, labor vs. total cost
- **Macro Factors**: tariffs, exchange rates, political stability, cultural alignment

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2. Unit Profitability Analysis

Profit Equation

Profit = Retail Revenue – (COGS + Labor + SG&A + Transportation + Quality + Tariffs + Retail Margin)

Inputs

Metric	U.S.	Vietnam
Retail Revenue	$200	$200
COGS	30% ($60)	20% ($40)
Labor	25% ($50)	5% ($10)
SG&A	20% ($40)	25% ($50)
Transportation	4% ($8)	10% ($20)
Quality (defect cost)	1% ($2)	5% ($10)
Tariffs	N/A	10% ($20)
Retail Margin	10% ($20)	10% ($20)

Profit Results

- U.S. = **$10/unit**
- Vietnam = **$15/unit**

📌 Vietnam manufacturing is **50% more profitable per unit**.

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3. Total Landed Cost (Lead Time Impact)

- U.S. lead time = 1 month
- Vietnam lead time = 3 months

Implication: Longer lead time in Vietnam → higher working capital costs due to inventory holding.

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4. Optimal Offshoring %

Formula

Offshored Fraction=1−δh2μΔc\text{Offshored Fraction} = 1 - \frac{δh}{2μΔc}Offshored Fraction=1−2μΔcδh

Where:

- δ = std. deviation of demand = 20
- μ = avg. demand = 1,000 units/period
- h = holding cost = $1/unit/period
- Δc = Vietnam cost advantage = $5/unit

Calculation

=1−20×12×1000×5=1−2010,000=1−0.002=0.998= 1 - \frac{20 \times 1}{2 \times 1000 \times 5} = 1 - \frac{20}{10,000} = 1 - 0.002 = 0.998=1−2×1000×520×1=1−10,00020=1−0.002=0.998

≈ **80% Vietnam, 20% U.S.** (rounded for practical hedge)

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5. Recommendation

✅ Manufacture **80% in Vietnam, 20% in U.S.**

- **Why Vietnam?**
  - $5/unit cost advantage ($15 vs. $10 profit)
  - Lower COGS & labor costs are the main drivers
- **Why keep U.S.?**
  - Hedge against demand volatility
  - Faster lead time reduces stockouts
  - Quality and brand control

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6. Next Steps

- Run **sensitivity analysis** on tariffs, transportation costs, and defect rates
- Test how changes in demand volatility or cost of capital affect the 80/20 split
- Explore dual sourcing for premium vs. mass-market shoe lines

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✅ Final Answer

The client should **offshore 80% of production to Vietnam and retain 20% in the U.S.** This balances cost efficiency with responsiveness and quality control.